

Jisun Hwang

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RESEARCH INTERESTS

Human-Computer Interaction	User-Centered Design	Large Language Models
Human-AI Collaboration	Data Analysis	Cues and Heuristic

EDUCATION

M.S. in Business Analytics, Ajou University, South Korea Thesis title: "Vibe Design: A Human-AI Collaborative Framework for the UI Design Process with Simulated Users" Thesis advisor: Prof. Juyoung Kang	Mar 2024 - Feb 2026
B.S in Living Design, Konkuk University, South Korea www.instagram.com/etang_endand	Mar 2017 - Feb 2022

PUBLICATIONS

Hwang, J., & Kang, J. (2026). Vibe Design: Human-in-the-loop AI Agents for UI Design with Large Language Models. In Proceedings of the 59th Hawaii International Conference on System Sciences, 4453-4462.

Choi, H., **Hwang, J.,** Lee, H., & Kang, J. (2026). LLM-Based Analysis of Small Business Complaints and Policy Demand Gaps. Journal of Intelligence and Information Systems, 32(1), 325-349.

Hwang, J., & Kang, J. (2025). Exploring Physical Store Customer Experience through BERTopic and Importance-Performance Analysis, Journal of Intelligence and Information Systems, 31(2), 85-110.

Hwang, J., Yoon, T., & Kang, J. (2025). An analysis of disability dispute for law improvement based on the BERT-LDA hybrid model. The Journal of Society for e-Business Studies, 30(1), 131-154.

Sun, I., Lee, D., Hwang, E., Kim, E., Kim, J., Yoo, J., Lee, A., Lee, M., Park, H., Kim, W., **Hwang, J.,** & Lee, H. (2025). Post-2030 global science, technology and innovation agenda study (Year 2): Deriving a global agenda-setting framework (Policy Research Report No. 2025-16-01). Science and Technology Policy Institute (STEPI).

RESEARCH EXPERIENCE

External Researcher Ajou University Business School "Multimodal Integration of EEG and Eye-Tracking for Establishing Consumer Reassurance Mechanisms in Intelligent Packaging: A Behavioral Immune System Perspective" with Prof. Cheul Rhee (funded by NRF: National Research Foundation of Korea) - Coordinated with an interdisciplinary research team on a three-year study of how intelligent packaging signals shape consumer reassurance and purchase behavior. - Explore EEG equipment options and build a custom eye-tracking apparatus for the study's neurophysiological measurements. - Develop and lead an independent study, "Selective Transfer of Expected Quality from Intelligent Packaging Cues in Online Grocery Shopping," examining how cue modality and third-party certification influence expected food quality across price levels.	Jul 2026 - Present
Graduate Researcher Ajou University Business School "A Study on Data-Driven Policy Demand Analysis and Improvement Tasks for Small Businesses" with Prof. Juyoung Kang (commissioned by SEMAS: Small Enterprise and Market Service) Established a demand-supply analysis framework to diagnose gaps between government support policies and the grievances of small business owners in Korea.	Aug 2025 - Dec 2025

- Applied topic modeling to online community and policy data, and built an LLM-based AI agent to match grievances with existing policies.
- Identified 30 grievance topics, including 10 not addressed by any existing policy; presented the findings to SEMAS and published them as a research paper.

External Researcher | Science and Technology Policy Institute (STePI)

Oct 2025 - Nov 2025

"From Agenda Following to Agenda Leadership: A Post-2030 Global STI Agenda Strategy (Year 2) and Agenda-Setting Framework" with **Prof. Hansol Lee** (commissioned by STePI)

- Applied hybrid topic modeling to national R&D project data (NTIS) to derive 80 AI research agendas.
- Detected 11 high-potential weak signals among the derived agendas and visualized them with a Keyword Issue Map.
- Co-authored the resulting government policy report as an external author.

Graduate Researcher | Ajou University Business School

Aug 2024 - Feb 2025

"An Analysis of Disability Disputes for Law Improvement" with **Prof. Juyoung Kang**

(funded by the National Research Foundation of Korea, Social Science Korea (SSK) Program)

- Collaborated with Prof. Taeyoung Yoon (School of Law) throughout the study, integrating legal expertise into topic labeling and the interpretation of dispute types.
- Applied a two-stage text-mining framework to news articles and legal consultations, deriving 6 dispute types in the guardianship system.
- Introduced the BERT-LDA hybrid topic model to Korean research for the first time, outperforming a BERT-only baseline in clustering quality.

PROJECTS

UI/UX Designer | WithData Inc.

Dec 2025 - Feb 2026

LG Electronics' Internal AI System Project

- Designed an identify-intervene debugging loop, informed by recent HCI research on agent debugging, enabling users to locate where chatbot outputs diverged from expectations and test corrections directly.
- Overhauled the chatbot's UI/UX to address user complaints about usability, structuring screen flows around familiar chat patterns and adding workflow features such as source file preview and product search.
- Delivered the full UI/UX design specification to secure the LG Electronics project contract.

Research Intern | UST Inc.

Jun 2025 - Aug 2025

Summer Industry-Academia Cooperation Program: Product Strategy for Truth of Beauty

- Collected consumer reviews across multiple web and mobile platforms, and applied BERT-LDA hybrid topic modeling to extract key consumer concerns.
- Conducted Importance-Performance Analysis (IPA) on rating data to identify 4 priority improvement areas, and analyzed 2017-2025 search trend data to track ingredient and formulation trends by product category.
- Proposed differentiation points for the brand's hero product based on the findings, and prepared and presented the final report.

Research Intern | Intellicode Inc.

Dec 2024 - Feb 2025

Winter Industry-Academia Cooperation Program: AI Agent for Automated Trending-Keyword Video Generation

- Developed the image generation component of a team-built AI agent that automatically produces videos from trending keywords.
- Engineered a diffusion-based image compositing pipeline in ComfyUI to flexibly combine virtual models with product images, integrated state-of-the-art image generation models, and prepared and presented the final report.

Research Intern | Nextorm Inc.

Jun 2024 - Aug 2024

Summer Industry-Academia Cooperation Program: Predictive Analysis of Tool Wear in CNC Processes

- Analyzed field sensor data from CNC processes to predict tool wear, aiming to replace operator judgment with data-driven detection.
- Trained a Random Forest model and applied SHAP values to quantify the influence of each variable.
- Identified data contamination from manual machine stoppages as the key barrier to statistically significant results, and compiled and presented the final report with recommendations for data collection.

WORK EXPERIENCE

Visual Merchandise Designer | Thursday Island by GNCO Co., Ltd.

Jan 2023 - Sep 2023

- Curated visual components that were adopted as the main visuals for the brand's official social media and website.
- Planned and designed visual displays for most of the brand's offline touchpoints, including in-store displays and pop-up stores, and supervised on-site execution.
- Developed weekly mannequin styling guidelines for new arrivals, coordinated implementation with store managers nationwide, and reported weekly sales of featured display products to the CEO.

3D Designer | RealBits Inc.

Apr 2022 - Dec 2022

AR Face-Tracking Avatar NFT Service

- Collaborated with a developer to launch a 3D character NFT service that let purchasers use their characters as AR face-tracking avatars across live streaming and messaging platforms.
- Conceived and modeled 3D characters and a range of accessories in Blender for Gen Z and millennial users.

TECHNICAL SKILLS

- Programming & Data Analysis: Python
- Computer aided design: Figma, Photoshop, Illustrator, Blender, ComfyUI
- Certifications: Big-Data Analysis Engineer (National Technical Qualification), Korea Data Agency

Dec 2025

AWARDS

- Outstanding Thesis Award (Dean's Award), Graduate School, Ajou University
- Best Presentation Paper Award, E-Bridge Society 2024 Fall Conference

Feb 2026

Oct 2024